

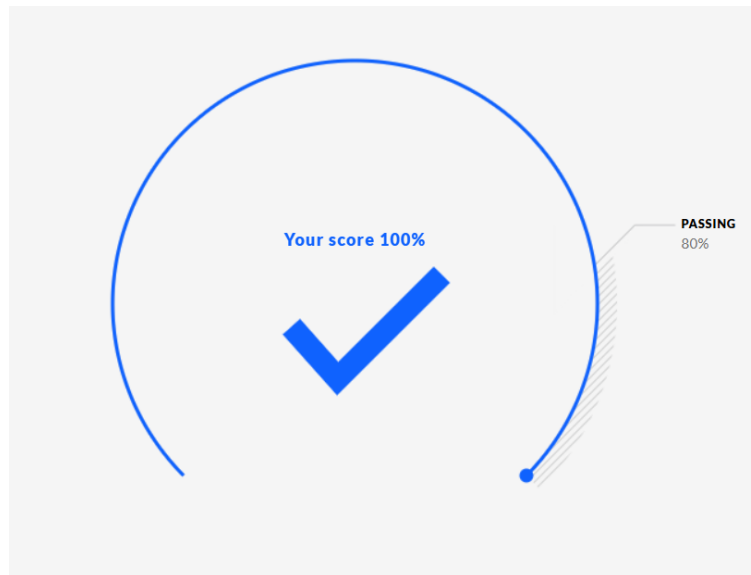
CURSO 3

MODULO 1

Open source software is software with code that is published publicly and can be used by anyone.

Open source software is collaborative, meaning it relies on a virtual community of people to review, change, and share source code with each other.

Developers share insights, ideas, and code to create more innovative software solutions.



MODULO 2

There are many useful data analysis tools and platforms! This module is designed to **introduce** you to common tools that data science professionals use to analyze and visualize data. You'll focus on:

- Microsoft Excel
- Google Sheets
- Structured Query Language (SQL)
- Python
- IBM Watson Studio
- Tableau
- Matplotlib

While this is not a complete list, it will orient you to the tools you'll learn about and can explore!

Working with Microsoft Excel

[Microsoft Excel](#) (MS Excel) is a spreadsheet application created by Microsoft. MS Excel is one of the most-used tools for data analysis.

You can use MS Excel to enter, examine, and interpret data in a variety of ways.

- You can manipulate and clean the data in rows and columns prior to analysis.
- MS Excel has built-in data analysis functions and features, such as filtering, formulas, and pivot tables.

You can also use MS Excel to help visualize data sets and gain insights from data.

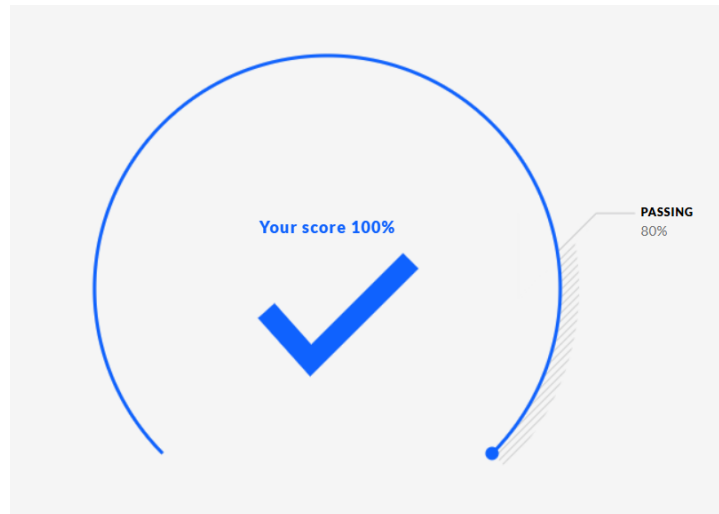
- MS Excel offers different types of charts, such as pie, bar, line, and scatter plot.
- You can use built-in templates and MS Excel's recommendations for a chart based on your data.
- You can combine different chart types on one spreadsheet in MS Excel.

Additional tools and programming languages

So far, you learned about a number of data analysis and visualization tools and some suggested resources if you would like to explore more.

Here are some additional data science technologies in the data science ecosystem.





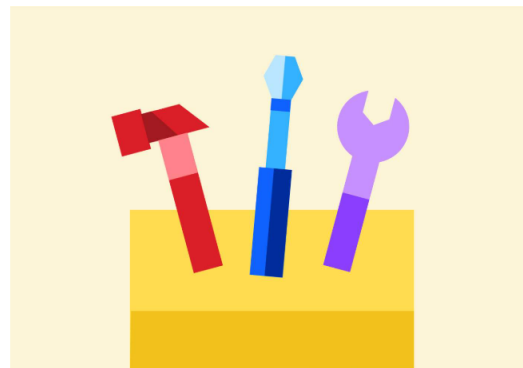
MODULO 3

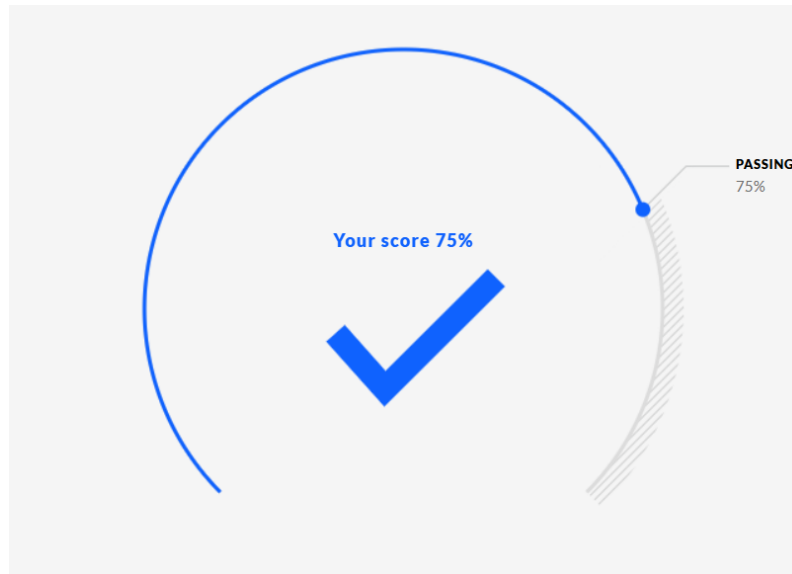
Your toolbox

You can *analyze* data with “**this**” tool or “**that**” programming language.

You can *visualize* data with “**that**” tool or “**this**” programming language.

Each data transformation tool is designed with a certain use in mind. Sometimes, tools overlap and serve the same purpose. But every tool has its advantages and disadvantages. The experiences can vary, and people will have their own preferences.





- 1 Open source software is software with code that is published publicly so anyone can use it and collaborate among a virtual community of people. Open source allows those working on data science projects to learn new software and develop new skills. It provides many potential benefits to businesses, such as accelerated innovation, cost savings, and competitiveness.
- 2 **GitHub** is a free, online service used to host, contribute, and collaborate on source code using repositories.
- 3 **Microsoft Excel** is software that's used for entering, manipulating, analyzing, and visualizing data. **Google Sheets** is a free, alternative to Microsoft Excel.
- 4 **Structured Query Language (SQL)** is a standard language to "query" or communicate with databases to modify them.
- 5 **IBM Watson Studio** has a data refinery tool with built-in charts and graphics to understand the distribution of data.
- 6 **Tableau** is a software for analyzing large volumes of data and creating different kinds of visualizations, including interactive dashboards.

Congratulations

You're almost done!

Earn a credential!

Reminder: This course is one of several in the Data Fundamentals series. **Complete all of the courses** in the Data Fundamentals learning plan and earn the **Data Fundamentals** digital credential from IBM!

